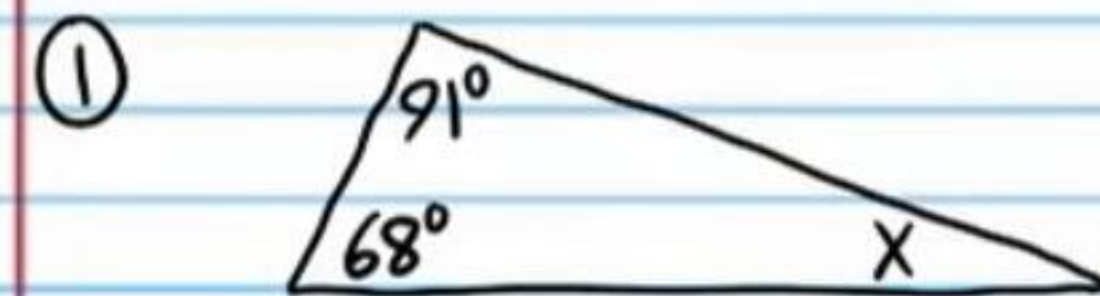
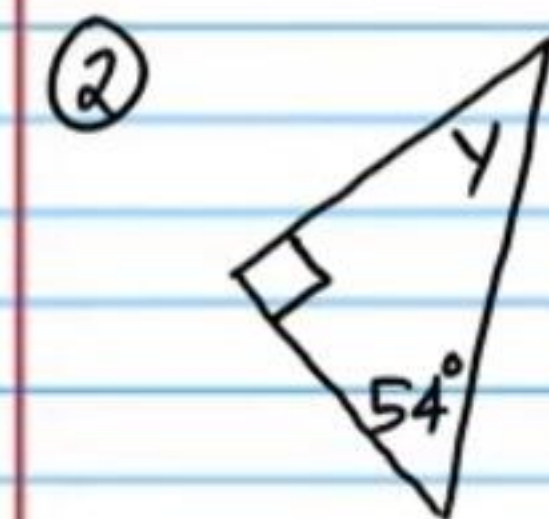


Geometry Review and the Six Basic Trigonometric Functions (Homework)

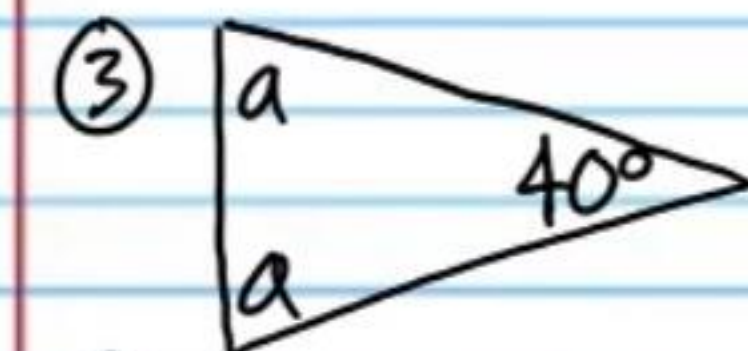
Find the value of each variable.



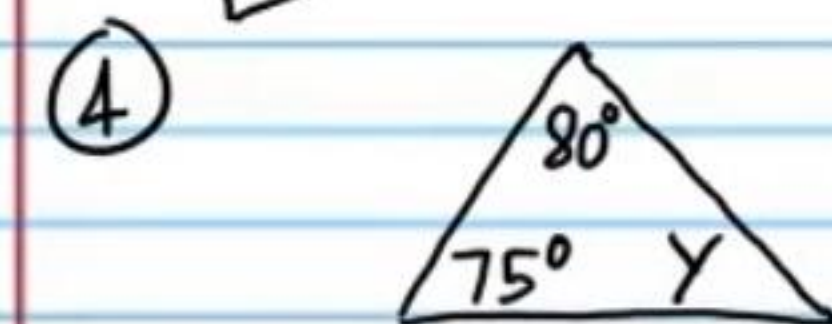
$$x = 21^\circ$$



$$y = 36^\circ$$

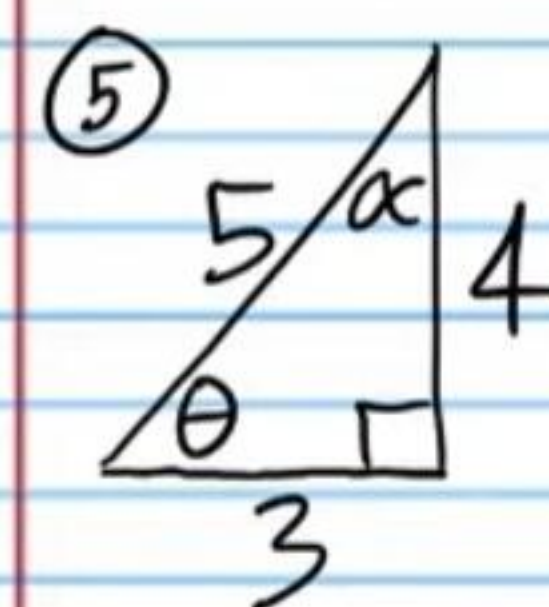


$$a = 70^\circ$$



$$y = 25^\circ$$

Find the values of the expressions below.



$$\sin \alpha \quad \csc \alpha$$

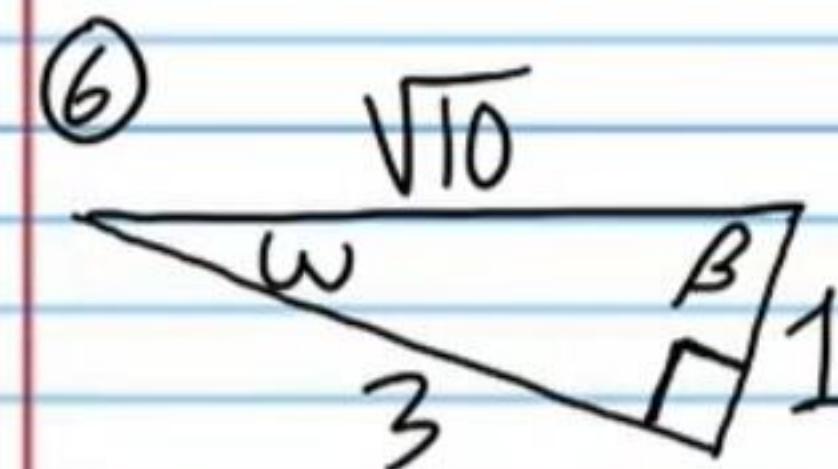
$$\cos \alpha \quad \sec \alpha$$

$$\tan \alpha \quad \cot \alpha$$

$$\sin \theta \quad \csc \theta$$

$$\cos \theta \quad \sec \theta$$

$$\tan \theta \quad \cot \theta$$



$$\sin w$$

$$\csc w$$

$$\cos w$$

$$\sec w$$

$$\tan w$$

$$\cot w$$

$$\sin \beta$$

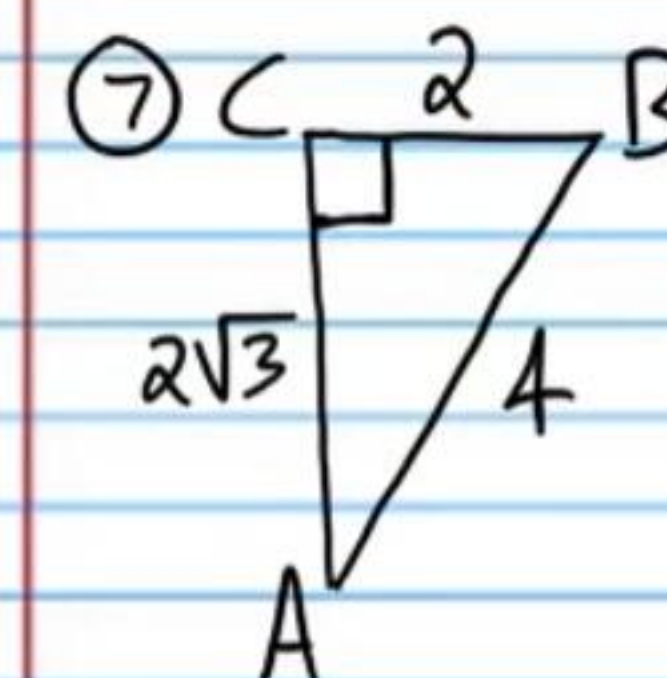
$$\csc \beta$$

$$\cos \beta$$

$$\sec \beta$$

$$\tan \beta$$

$$\cot \beta$$



$$\sin A$$

$$\csc A$$

$$\cos A$$

$$\sec A$$

$$\tan A$$

$$\cot A$$

$$\sin B$$

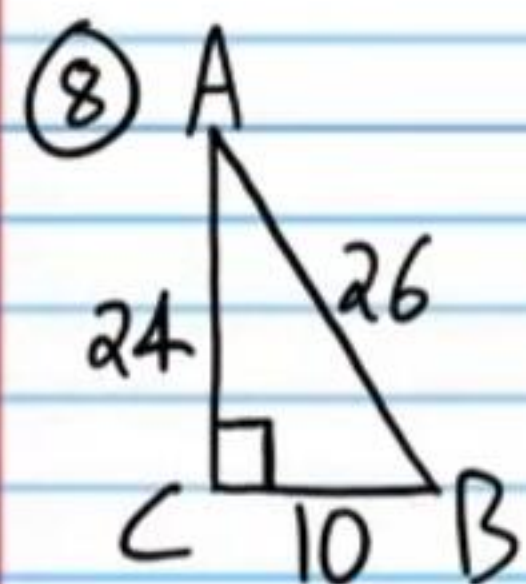
$$\csc B$$

$$\cos B$$

$$\sec B$$

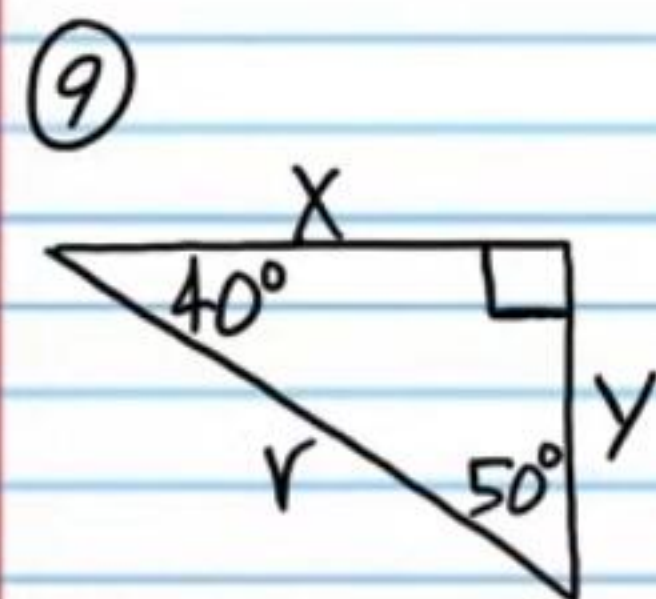
$$\tan B$$

$$\cot B$$

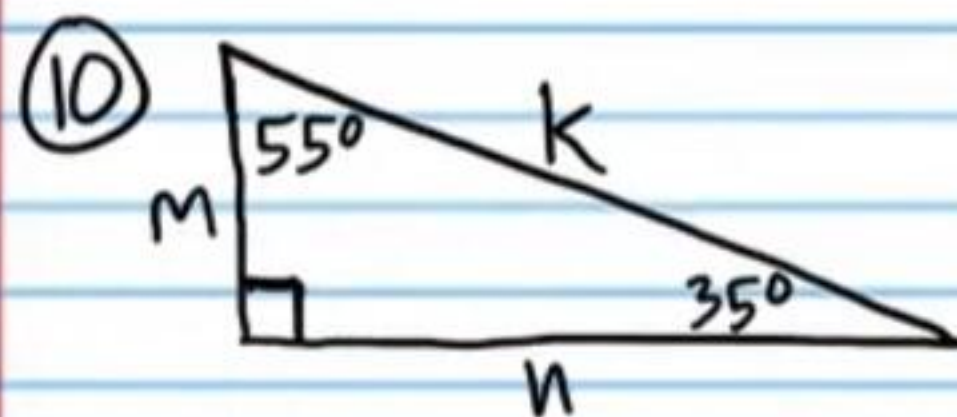


$\sin A$	$\csc A$
$\cos A$	$\sec A$
$\tan A$	$\cot A$
$\sin B$	$\csc B$
$\cos B$	$\sec B$
$\tan B$	$\cot B$

Find an exact ratio for each expression listed below.



$\sin 40^\circ$	$\csc 40^\circ$
$\cos 40^\circ$	$\sec 40^\circ$
$\tan 40^\circ$	$\cot 40^\circ$
$\sin 50^\circ$	$\csc 50^\circ$
$\cos 50^\circ$	$\sec 50^\circ$
$\tan 50^\circ$	$\cot 50^\circ$



$\sin 55^\circ$	$\csc 55^\circ$
$\cos 55^\circ$	$\sec 55^\circ$
$\tan 55^\circ$	$\cot 55^\circ$
$\sin 35^\circ$	$\csc 35^\circ$

$\cos 35^\circ$

$\sec 35^\circ$

$\tan 35^\circ$

$\cot 35^\circ$

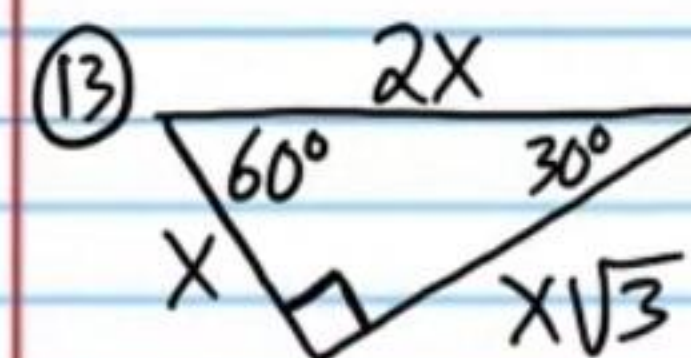
Use a calculator to find the sine, cosine, and tangent of each of the following angles. Round to four decimal places.

★ Be sure that your calculator is in degree mode, not radian mode.

⑪ $\theta = 68^\circ$

⑫ $w = 15^\circ$

Find the value of each expression below without a calculator.



$\sin 60^\circ$

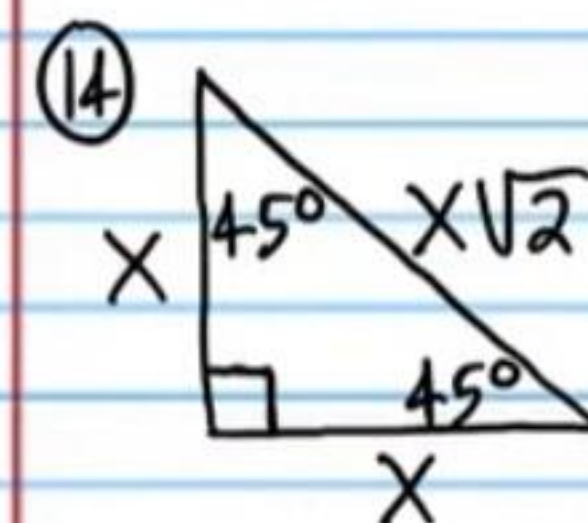
$\csc 30^\circ$

$\cos 60^\circ$

$\sec 60^\circ$

$\tan 30^\circ$

$\cot 30^\circ$



$\sin 45^\circ$

$\cos 45^\circ$

$\tan 45^\circ$

$\csc 45^\circ$

$\sec 45^\circ$

$\cot 45^\circ$

Answers for #'s 5-14

$$\textcircled{5} \quad \sin \alpha = \boxed{\frac{3}{5}} \quad \csc \alpha = \boxed{\frac{5}{3}}$$

$$\cos \alpha = \boxed{\frac{4}{5}} \quad \sec \alpha = \boxed{\frac{5}{4}}$$

$$\tan \alpha = \boxed{\frac{3}{4}} \quad \cot \alpha = \boxed{\frac{4}{3}}$$

$$\sin \theta = \boxed{\frac{4}{5}} \quad \csc \theta = \boxed{\frac{5}{4}}$$

$$\cos \theta = \boxed{\frac{3}{5}} \quad \sec \theta = \boxed{\frac{5}{3}}$$

$$\tan \theta = \boxed{\frac{4}{3}} \quad \cot \theta = \boxed{\frac{3}{4}}$$

$$\textcircled{6} \quad \sin w = \boxed{\frac{\sqrt{10}}{10}} \quad \csc w = \boxed{\sqrt{10}}$$

$$\cos w = \boxed{\frac{3\sqrt{10}}{10}} \quad \sec w = \boxed{\frac{\sqrt{10}}{3}}$$

$$\tan w = \boxed{\frac{1}{3}} \quad \cot w = \boxed{3}$$

$$\sin \beta = \boxed{\frac{3\sqrt{10}}{10}} \quad \csc \beta = \boxed{\frac{\sqrt{10}}{3}}$$

$$\cos \beta = \boxed{\frac{\sqrt{10}}{10}} \quad \sec \beta = \boxed{\sqrt{10}}$$

$$\tan \beta = \boxed{3}$$

$$\cot \beta = \boxed{\frac{1}{3}}$$

$$\textcircled{7} \quad \sin A = \boxed{\frac{1}{2}}$$

$$\csc A = \boxed{2}$$

$$\cos A = \boxed{\frac{\sqrt{3}}{2}}$$

$$\sec A = \boxed{\frac{2\sqrt{3}}{3}}$$

$$\tan A = \boxed{\frac{\sqrt{3}}{3}}$$

$$\cot A = \boxed{\sqrt{3}}$$

$$\sin B = \boxed{\frac{\sqrt{3}}{2}}$$

$$\csc B = \boxed{\frac{2\sqrt{3}}{3}}$$

$$\cos B = \boxed{\frac{1}{2}}$$

$$\sec B = \boxed{2}$$

$$\tan B = \boxed{\sqrt{3}}$$

$$\cot B = \boxed{\frac{\sqrt{3}}{3}}$$

$$\textcircled{8} \quad \sin A = \boxed{\frac{5}{13}}$$

$$\csc A = \boxed{\frac{13}{5}}$$

$$\cos A = \boxed{\frac{12}{13}}$$

$$\sec A = \boxed{\frac{13}{12}}$$

$$\tan A = \boxed{\frac{5}{12}}$$

$$\cot A = \boxed{\frac{12}{5}}$$

$$\sin B = \boxed{\frac{12}{13}}$$

$$\csc B = \boxed{\frac{13}{12}}$$

$$\cos B = \boxed{\frac{5}{13}}$$

$$\sec B = \boxed{\frac{13}{5}}$$

$$\tan B = \frac{12}{5}$$

$$\cot B = \frac{5}{12}$$

$$\textcircled{9} \sin 40^\circ = \frac{y}{r}$$

$$\csc 40^\circ = \frac{r}{y}$$

$$\cos 40^\circ = \frac{x}{r}$$

$$\sec 40^\circ = \frac{r}{x}$$

$$\tan 40^\circ = \frac{y}{x}$$

$$\cot 40^\circ = \frac{x}{y}$$

$$\sin 50^\circ = \frac{x}{r}$$

$$\csc 50^\circ = \frac{r}{x}$$

$$\cos 50^\circ = \frac{y}{r}$$

$$\sec 50^\circ = \frac{r}{y}$$

$$\tan 50^\circ = \frac{x}{y}$$

$$\cot 50^\circ = \frac{y}{x}$$

$$\textcircled{10} \sin 55^\circ = \frac{n}{k}$$

$$\csc 55^\circ = \frac{k}{n}$$

$$\cos 55^\circ = \frac{m}{k}$$

$$\sec 55^\circ = \frac{k}{m}$$

$$\tan 55^\circ = \frac{n}{m}$$

$$\cot 55^\circ = \frac{m}{n}$$

$$\sin 35^\circ = \frac{m}{k}$$

$$\csc 35^\circ = \frac{k}{m}$$

$$\cos 35^\circ = \frac{n}{k}$$

$$\sec 35^\circ = \frac{k}{n}$$

$$\tan 35^\circ = \frac{m}{n}$$

$$\cot 35^\circ = \frac{n}{m}$$

$$\textcircled{11} \sin 68^\circ \approx 0.9272$$

$$\cos 68^\circ \approx 0.3746$$

$$\tan 68^\circ \approx 2.4751$$

$$\textcircled{12} \sin 15^\circ \approx 0.2588$$

$$\cos 15^\circ \approx 0.9659$$

$$\tan 15^\circ \approx 0.2679$$

$$\textcircled{13} \sin 60^\circ = \frac{\sqrt{3}}{2}$$

$$\csc 30^\circ = 2$$

$$\cos 60^\circ = \frac{1}{2}$$

$$\sec 60^\circ = 2$$

$$\tan 30^\circ = \frac{\sqrt{3}}{3}$$

$$\cot 30^\circ = \sqrt{3}$$

$$\textcircled{14} \sin 45^\circ = \frac{\sqrt{2}}{2}$$

$$\csc 45^\circ = \sqrt{2}$$

$$\cos 45^\circ = \frac{\sqrt{2}}{2}$$

$$\sec 45^\circ = \sqrt{2}$$

$$\tan 45^\circ = 1$$

$$\cot 45^\circ = 1$$